

HEC-RAS RUN SHEET - WADI MAJLAS DAM BREAK ANALYSIS

Companion to the Dam Break Analysis Methodology Report Rev 00 (September 2026). Values only; the reasoning is in Rev 00, chapter given in brackets. Units: m, m a.s.l., m³/s, h. Project CRS EPSG:32640. HEC-RAS 6.6, SI units. Levels: wadi bed 23.0, FSL 84.5, crest 95.5, parapet 96.7, PMF pool 96.7.

1.1. WHAT TO WRITE DOWN

Table 1 - Run register.

Run register column	Entry
Run ID	S1-D, S1-N, S2-D, S2-N, S3-D, S3-N; sensitivity SW1, SW2, SW3, ST1, ST2, SC1, SN1, SN2, SM1, SS1, SP1, SF1, SI1, SR1, SPW1
HEC-RAS files	Project Majlas_DamBreak.prj; plan = Run ID; geometry G_Dikes or G_NoDikes; unsteady U_S1 (no inflow), U_PMF, U_10000; terrain T_Dikes or T_NoDikes
Inputs	Start pool; inflow file; breach width, bottom, time, coefficient, trigger; sea level; dikes; mesh version; Manning table version
Run	Date, modeller, HEC-RAS version, run time, restart file used
Checks	Volume error %; max Courant; cells with iteration warnings; peak within hand-check bounds (Y/N); reviewer initials

Table 2 - Results to record.

Value to record from every run	Where
Breach peak outflow and its time; breach volume	Connection hydrograph (10 s output)
Reservoir level at breach start, at peak outflow, 1 h and 6 h after	Reservoir 2D area stage at the dam
Peak flow, peak level, arrival time (depth 0.3 m), time to peak	Reference lines: gorge exit, town entry, coast road, shoreline
Maximum depth, velocity, D×V at 10 named points in the town	RAS Mapper point values
Inundated area above 0.3 m (km ²); people in it; plots in it	Max depth raster > 0.3 m, overlay GHS-POP and land-use
Volume accounting error; max Courant; iteration warnings; run time	Computation log, HDF summary
Exported rasters (2 m GeoTIFF)	Max depth, max velocity, max D×V, arrival time 0.3 m, duration > 0.3 m, hazard AIDR; incremental depth for S2 (S2 minus S3)

1.2. MODEL SET-UP

Table 3 - Geometry.

Step	Do this	Value
Terrain	RAS Mapper > Terrains > New; stack in this order, top has priority	1 dam-site survey 5 cm; 2 design surfaces (dam, spillway, basin, chute, dikes, crossings, roads); 3 drone DTM 2 m; 4 NSA 5 m (existing DS model); 5 NSA 2 m
Two terrains	Repeat without the dike surfaces	T_Dikes; T_NoDikes



Step	Do this	Value
Dam in terrain	Cut the dam body out along the axis (terrain modification, channel) so the connection alone represents it	Bed at axis 23.0 m
Land cover	Map Layers > Land Cover from the land-use layer; Manning table	See table below
Reservoir 2D area	New 2D area from the dam to the upstream end of the 96.7 m pool plus 200 m margin	Cells 30 to 50 m; 10 m in the last 1 km; check volume at FSL = 102.2 Mm ³ ± 2 %
Downstream 2D area	Reuse 'DS Protection' from the flood-protection model; extend to the shoreline; add the sea strip	Cells 10 to 15 m plain; 5 m dam and basin; 10 m town; 20 to 30 m coast
Breaklines	Dikes (both crests), wadi banks, road embankments, coast road, dam axis, basin walls	Cell size along breakline = zone value; enforce
Refinement regions	Dam and basin 5 m; town 10 m	-
Reference lines	Gorge exit, town entry, coast road, shoreline	Named as listed
Boundary condition lines	Reservoir inflow (upstream end); tributary inflows at HEC-HMS junctions J10, J11, J12; sea (stage); open land edges (normal depth)	-

Table 4 - Manning's n.

Land cover	n base	n low	n high
Wadi bed, gravel	0.035	0.030	0.045
Floodplain, bare or scrub	0.040	0.035	0.050
Agriculture, date gardens	0.080	0.060	0.100
Residential and commercial blocks	0.120	0.080	0.200
Roads, paved	0.018	0.016	0.022
Concrete (spillway, basin)	0.015	0.013	0.017
Riprap on dikes	0.035	0.030	0.040
Reservoir bed	0.035	0.030	0.045
Sea, lagoon	0.025	0.020	0.030

Table 5 - Dam connection.

SA/2D connection field	Entry
Alignment	Straight dam axis, 312 m, drawn from the design CAD (indicative shapefile in 04_data/gis until the CAD arrives)
Weir profile (station, elevation)	Left non-overflow blocks: 95.5; spillway section 212 m gross: 5 bays × 40 m at 84.5 with 4 piers × 3 m at 96.7; right non-overflow blocks: 95.5. Block stations from the joint layout when received; until then left 0-50 m, spillway 50-262 m, right 262-312 m
Weir width	6 m
Weir shape, non-overflow blocks	Broad crested; coefficient 1.66 (SI)
Weir shape, spillway bays	Ogee; spillway approach height 61.5 m; design energy head from the spillway design (request); resulting C about 2.1 to 2.2 (SI)
Culverts, gates	None (bottom outlet closed)
Check	No-failure run S3-D: discharge at pool 96.7 m = 17,000 m ³ /s ± 5 %



Table 6 - Breach entries.

Plan > Breach (plan data)	Base	Lower	Upper	Other sensitivity
Breach this structure	On (S1, S2)	On	On	Off in S3
Breach method	User entered data	-	-	-
Center station	Centre of the spillway section	-	-	-
Final bottom width (m)	86	43	156	129 (three bays)
Final bottom elevation (m)	23.0	-	-	-
Left and right side slopes (H:V)	0	0	0	-
Breach weir coefficient (SI)	1.66	1.44	1.66	-
Full formation time (h)	0.1	0.3	0.05	-
Failure mode	Overtopping	-	-	-
Trigger, S1 (sunny day)	Set time = simulation start + 2 h	-	-	-
Trigger, S2 (flood day)	WS elevation = 96.7 m	-	-	SF1: set time = time of max pool in S3-D + 2 h
Breach progression	Linear	-	-	-

1.3. SCENARIOS AND WATER LEVELS

Table 7 - Base runs.

Run	Geometry / terrain	Reservoir initial WSE (m)	Inflow BC	Tributaries J10-J12	Breach	Sea stage	Window
S1-D	G_Dikes / T_Dikes	84.5	None	None	On; set time +2 h; 86 m; 0.1 h	HAT, constant	0 to 14 h; warm-up 2 h; restart written at 2 h
S1-N	G_NoDikes / T_NoDikes	84.5	None	None	Same as S1-D	HAT	Same
S2-D	G_Dikes / T_Dikes	84.5	PMF hydrograph, 5-min, 48 h	PMF local runoff	On; WS elev 96.7 m; 86 m; 0.1 h	HAT	0 to 60 h
S2-N	G_NoDikes / T_NoDikes	84.5	PMF	PMF	Same as S2-D	HAT	0 to 60 h
S3-D	G_Dikes / T_Dikes	84.5	PMF	PMF	Off	HAT	0 to 60 h; run first; checks before S1/S2
S3-N	G_NoDikes / T_NoDikes	84.5	PMF	PMF	Off	HAT	0 to 60 h

Table 8 - Water levels.

Level	m a.s.l.	Used as
Wadi bed at the axis; breach bottom	23.0	Final bottom elevation
Bottom outlet intakes	36 and 44	Closed; SP1 starts the pool at 46
FSL = ogee crest	84.5	Initial WSE of every run; spillway weir elevation
Non-overflow crest	95.5	Weir elevation of the blocks
Parapet top and pier top	96.7	Pier elevation; SPW1 uses it for the blocks too



Level	m a.s.l.	Used as
PMF pool (to be confirmed: 96.0 in the design table)	96.7	Breach trigger in S2; check level of S3 $\pm$ 0.2 m
Sea: highest astronomical tide (HAT)	from tide tables	Sea stage, all base runs
Sea: mean sea level	0.0	SS1
Spillway capacity	17,000 m ³ /s	S3-D discharge at the PMF pool $\pm$ 5 %

Table 9 - Sensitivity runs.

ID	Copy of	Change	Value
SW1	S1-D, S2-D	Final bottom width	43 m
SW2	S1-D, S2-D	Final bottom width	129 m
SW3	S1-D, S2-D	Final bottom width	156 m
ST1	S1-D, S2-D	Formation time	0.3 h
ST2	S1-D, S2-D	Formation time	0.05 h
SC1	S1-D	Breach weir coefficient	1.44
SN1	S1-D, S2-D	Manning table	n low column
SN2	S1-D, S2-D	Manning table	n high column
SM1	S1-D	Mesh	Half the cell size at the dam and in the town
SS1	S2-D	Sea stage	Mean sea level 0.0
SP1	S1-D	Reservoir initial WSE	46 m
SF1	S2-D	Trigger	Set time = time of max pool in S3-D + 2 h
SI1	S2-D	Inflow	10,000-year hydrograph
SR1	S1-D	Reservoir	Storage area with the elevation-volume curve instead of the 2D area
SPW1	S3-D	Weir profile	Non-overflow blocks at 96.7 m (parapet credited)

1.4. COMPUTATION, OUTPUTS AND CHECKS

Table 10 - Computation and output settings.

Where	Setting	Value
Unsteady Flow > Boundary Conditions	Reservoir BC line	Flow hydrograph: PMF (S2, S3), 10,000-yr (SI1), none (S1)
	Tributary BC lines J10, J11, J12	Flow hydrographs from HEC-HMS (PMF local runoff); none in S1
	Sea BC line	Stage hydrograph, constant HAT (SS1: 0.0)
	Open land edges	Normal depth, friction slope from the terrain at the edge
Unsteady Flow > Initial Conditions	Reservoir 2D area	Initial WSE 84.5 m (SP1: 46 m); downstream area dry
Plan > Computation Options > 2D	Equation set	Shallow Water Equations, Eulerian-Eulerian (SWE-EM)
	Theta (implicit weighting)	1.0
	Water surface tolerance / volume tolerance	0.003 m / 0.003 m
	Maximum iterations	20
	Turbulence	Conservative, default coefficients
Plan > Advanced Time Step	Time step	Adaptive on Courant: max 1.0, halve above; min 0.5, double below; initial 1 s



Where	Setting	Value
Plan > Simulation window	S1	0 to 14 h (breach at 2 h)
	S2, S3	0 to 60 h
Plan > Output	Hydrograph output interval	10 s
	Mapping output interval	1 min for 2 h after the breach, then 5 min
	Detailed output; restart file	On; write restart at 2 h in S1-D and reuse for S1 sensitivity runs
RAS Mapper > Results	Export	Max depth, max velocity, max D×V, arrival time (0.3 m), duration (> 0.3 m), hazard AIDR; GeoTIFF 2 m, EPSG:32640

Table 11 - Checks.

Check	Pass if	Run
Reservoir volume	2D area volume at 84.5 m within 2 % of the elevation-volume curve	Before any run
Spillway rating	S3-D discharge at the PMF pool within 5 % of 17,000 m ³ /s	S3-D
Routed pool	S3-D peak pool within 0.2 m of the design team's value	S3-D
Volume error	Below 1 %	Every run
Stability	No oscillation in the breach hydrograph; no iteration warnings at the peak; smooth arrival-time map	Every run
Time step	Result unchanged when max Courant is halved	S1-D, S2-D
Mesh	Peak depth at the town entry changes less than 5 % in SM1	SM1
Peak outflow	Between the Ritter and the weir hand values: S1 base 38,000 to 71,000; S2 base 50,000 to 93,000 m ³ /s, or explained	Every failure run
Water reaches the sea	No volume lost at open edges; sea outflow plus storage equals inflow plus breach volume	Every run
Counts	People and plot counts reproduced by an independent QGIS overlay	Base runs

